

ROCHISH G S K

APPLIED AI SCIENTIST

Hyderabad, | +918106833102 | rochish.gsk@outlook.com

Professional summary

Applied AI Scientist with expertise in machine learning, deep learning, generative AI, Graph Neural Networks, and quantum machine learning. Experienced in developing end-to-end AI solutions using PyTorch, TensorFlow, and Python, with a strong background in research, model development, and deploying AI for real-world scientific applications.

Work experience

Research Scientist (AI Developer) | PharmCADD Korea | Sep 2021 – Present

- Developed and customized GNN-based structure aware molecular generation models for de novo drug discovery using 3D protein pocket information.
- Built hybrid classical–quantum machine learning models for molecular generation using QCBMs and deep learning architectures.
- Designed and implemented protein–protein interaction prediction models using deep learning and sequence-based feature representations.
- Developed LSTM-GRU reinforcement learning models for generating novel drug-like molecules.
- Built ensemble machine learning models using XGBoost, Random Forest, and Deep Neural Networks to improve molecular property prediction accuracy.
- Developed deep learning models for RNA small-molecule prediction and biological sequence analysis.
- Performed molecular data preprocessing, feature engineering, visualization, and statistical analysis using Python, RDKit, Pandas, and NumPy.
- Collaborated with multidisciplinary research teams to integrate AI solutions into computational drug discovery workflows.

Machine Learning Engineer | Sai Life Sciences Ltd , Hyderabad | Apr 2021 – Sep 2021

- Developed machine learning and deep learning models for predicting molecular properties including half-life, solubility, Caco-2 permeability, bioavailability, CYP2D6 inhibition, LD50, and hepatocyte clearance.
- Built predictive QSAR models using Python, RDKit, XGBoost, and deep learning techniques.
- Performed molecular data preprocessing, feature engineering, and exploratory data analysis on large cheminformatics datasets.
- Applied attention-based transformer architectures for molecular property prediction.
- Created similarity maps and molecular visualizations to interpret model predictions and identify important chemical substructures.
- Collaborated with research scientists to support AI-driven drug discovery projects.

AI Chatbot (Personal Project) | Freelance | Jan 2022 – Sep 2022

- Designed and developed an intelligent NLP chatbot capable of processing both text and speech queries.
- Implemented speech recognition and text-to-speech functionality for conversational interactions.
- Built the chatbot using Python, Flask, NLTK, and PyTorch.
- Integrated the chatbot into a product-based website to automate customer support.

Education

PGDM | IIIT - B | Sep 2022 - Sep 2023

B.Tech | JNTU-H | Jun 2014 - May 2018

Skills

- Python
- Quantum Machine Learning
- Transformers
- Numpy
- Seaborn
- PennyLane
- Reinforcement Learning
- GNN
- Git
- QCBM
- PyTorch
- Tensorflow
- Data analytics
- Pandas
- Qiskit
- Quantum Computing
- RDkit
- LSTM/GRU
- VQC

Courses

IIT Roorke | Data Science Specialisation | Sep 2021 – Sep 2022

IIT Roorke | Deep Learning | Aug 2021 – Aug 2021

IIT Roorke | Big Data Engineering | Aug 2021 – Aug 2021

Links

[LinkedIn](#)

[Github](#)

Internships

Machine Learning Intern Sai Life Sciences ltd | Dec 2020 – Mar 2021

Developed a machine learning model for predicting blood-brain barrier permeability using molecular descriptors.

Performed adverse event data analysis and preprocessing to support predictive modeling.

Automated data preparation and visualization workflows using Python.